MILITARY SPECIFICATION SHEET

ELECTRON TUBE, CATHODE RAY

TYPE 5BGP2

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Electrostatic deflection and focus, flat face, aluminized screen

DIMENSIONS AND PIN CONNECTIONS: See figure 1

ABSOLUTE RATINGS:

Parameter:	Ef	Ec1	ed	Eb1	E defl plate avg	Eb2
Unit:	V	Vdc	vdc	Vdc	Vdc	Vdc
Maximum:	6.9	0	550	880	2,200	2,300
Minimum:	5.7	-200	---	---	---	---
TEST CONDITIONS: (see note 1)	6.3	Adjust	---	Focus (see note 2)	1,650	Adjust (see note 2)

ABSOLUTE RATINGS:

Parameter:	Eb3	Rg	Zd	Eb3/E defl plate avg	Ehk	Eis
Unit:	Vdc	Meg	Meg	---	Vdc	Vdc
Maximum:	13,200	1.5	1.5	6.1	±180	2,300
Minimum:	6,000	---	---	5.9	---	---
TEST CONDITIONS: (see note 1)	10,000	---	---	---	---	Adjust for minimum pattern distortion (see note 3)

GENERAL:

(A) Qualification - Not required

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T15:12Z
Check the source to verify that this is the current version before use.

METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1							
5201	Voltage breakdown		---	---	---	---	---	---
5201	Voltage breakdown (electrostatic types)		---	---	---	---	---	---
5106	Bulb, screen, and faceplate quality		---	---	---	---	---	---
5223	Modulation	Ib3 = 25 µAdc	---	---	ΔEc	---	50	Vdc
5231	Spot position (electro-static deflection)		---	---	---	---	10	mm
5241	Grid-cutoff voltage		---	---	Eco	-50	-80	Vdc
5221	Light output	Ib3 = 25 µAdc	---	---	---	175	---	ftL
5206	Gas "cross"	Ib3 = 25 µAdc	---	---	---	---	---	---
5251	Grid No. 1 leakage current		---	---	---	---	---	---
5251	Anode No. 2 leakage current		---	---	---	---	---	---
---	Isolation shield voltage (Eis)	See note 7	---	---	Eis	1,600	1,700	Volts
---	Spot size	See note 8	---	---	---	---	0.028	in.
---	Deflection defocusing	See note 5	---	---	---	---	---	---
5251	Anode No. 1 leakage current		---	---	---	---	---	---
---	Deflection-plate shading	Centered with respect to tube face (see note 4)	---	---	---	---	---	---
		1D2	---	---	Ib3	5.0	---	µAdc
		3D4	---	---	Ib3	5.0	---	µAdc
5103	Pattern distortion	See note 3	---	---	---	---	---	---
	Quality conformance inspection, part 2							
1301	Heater current		---	---	If	540	660	mA
5201	Electrode current (anode No. 1)	Ec1 = 0	---	---	Ib1	-15	10	µAdc
5201	Electrode current (cathode)	Ib3 = 25 µAdc	---	---	Ik	---	650	µAdc
5246	Focusing voltage at cutoff		---	---	Eb1	300	450	Vdc
5251	Heater-cathode leakage		---	---	---	---	---	---
1101	Secureness of base, cap, or insert		---	---	---	---	---	---
5248	Deflection factor	1D2	---	---	DF	70.0	86.0	Vdc/in.
5248	Deflection factor	3D4	---	---	DF	28.4	34.8	Vdc/in.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T15:12Z
Check the source to verify that this is the current version before use.

METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 2 -Continued							
5201	Electrode current (anode No. 3)	See note 6	---	---	---	5.6	42.0	μAdc
5101	Angle between traces		---	---	---	89	91	Degrees
5101	Base alinement (electrostatic types)	+3D4: pin No. 1	---	---	---	---	---	---
5101	Side terminal alinement	+1D2	---	---	---	---	5	Degrees
5101	Neck and base alinement (electrostatic types)		---	---	---	---	2	Degrees
5246	Focusing voltage at modulation condition		---	---	Eb1	250	450	Vdc
1105	Permanence of marking		---	---	---	---	---	---
	Quality conformance inspection, part 3							
---	Life-test provisions	Group C; Ib3 = 15 μAdc; Eb3 = 10 kVdc; Eis = Eb2 = 1,650 Vdc; t = 500 hours (min)	---	---	---	---	---	---
---	Life-test end points							
	Modulation		---	---	ΔEc1	---	50	Vdc
	Grid 1 leakage current		---	---	---	---	---	---
	Anode 2 leakage current		---	---	---	---	---	---
	Heater-cathode leakage current		---	---	---	---		
	Periodic-check tests (see note 9)							
1216	Base material insulating quality	Zone 5 (minimum)	---	---	---	---	---	---
1141	Pressure (implosion)		---	---	---	---	---	---
5101	Neck and bulb alinement (electrostatic types)		---	---	---	---	2.25	in.
5101	Face tilt		---	---	---	---	0.042	in.
1331	Direct-interelectrode capacitance	k to all	---	---	Ck	---	5.4	pF
		g1 to all	---	---	Cg1	---	8.9	pF
		D1 to D2	---	---	C1D2	---	2.0	pF
		D3 to D4	---	---	C3D4	---	1.4	pF
		D1 to all	---	---	CD1	---	6.0	pF
		D2 to all	---	---	CD2	---	6.0	pF
		D3 to all	---	---	CD3	---	4.5	pF
		D4 to all	---	---	CD4	---	4.5	pF
5111	Vibration		---	---	---	---	1.0	mm
Ⓐ 5115	Shock		---	---	---	---	---	---
5248	Deflection-factor uniformity		---	---	---	---	---	---

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T15:12Z
Check the source to verify that this is the current version before use.

METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
						MIN	MAX	
	Periodic-check tests (see note 9) - Continued							
5216	Stray light emission (conventional types)		---	---	---	---	---	---
5216	Cathode illumination		---	---	---	---	---	---

NOTES:

1. All readings of beam current (Ib3) shall be in addition to the reading obtained for anode No. 3 current in quality conformance inspection, part 1.

2. Under the operating conditions listed, unless otherwise specified, Eb2 and Eb1 shall be adjusted for optimum center screen focus. Under this condition, the value of Eb2 shall be within the limits of 1,585 to 1,830 volts. To maintain proper astigmatism adjustment as total cathode current is varied, resistance in the accelerator circuit should not exceed 5,000 ohms.

3. Adjust the isolation shield voltage (Eis) for optimum 60 by 100 mm raster. The center of the raster shall be coincident with the tube center. The 60 by 100 mm area thus defined is considered the useful scan area of the tube and is used in all subsequent applicable tests. The raster edges shall not deviate from straight parallel lines by more than 1.0 mm total on left and right edges, nor by more than 1.0 mm total on top and bottom edges.

   Without further adjustments made on the isolation electrode, the above test shall be repeated, using a 60 mm vertical trace and 100 mm horizontal trace, respectively. The vertical and horizontal traces shall be positioned to the edges of the 60 by 100 mm area. This test shall be performed after a 100 mm horizontal trace of normal intensity has been present on the tube face for at least 5 minutes. If (Eis) must be adjusted to optimize either deflection factor, deflection-factor uniformity, or useful scan test, all tests including pattern distortion tests shall be repeated at the new value of Eis.

4. With the tube setup and adjusted as outlined in note 3, the voltages on deflection plates are adjusted to produce a single horizontal trace 10 cm in length. The beam current (Ib3) is adjusted to give a current of 10 μA when the trace is centered as outlined in note 3. With no further adjustment of the beam current, the 10-cm long trace is positioned 3 cm above and then 3 cm below the vertical center of the useful scan area. At the full excursion of 3 cm above and 3 cm below the vertical center of the useful scan area, the beam current (Ib3) shall not be less than the limits shown on the tube specification sheet.

   The deflection plate voltages are now adjusted to produce a single vertical trace 6-cm long, centered as outline in note 3 and the beam current (Ib3) adjusted to 10 μA. With no further adjustment of beam current, the trace shall then be positioned first 5 cm to the right and then 5 cm to the left of the center of the useful scan area. At the full excursion of 5 cm right and 5 cm left, from the center of the useful scan area, the beam current (Ib3) shall not be less than the value specified on the tube specification sheet.

5. Deflection defocusing shall be measured with focus, astigmatism, and intensity unchanged from that determined for measurement of spot size. With the spot located anywhere within the useful scan area, its maximum dimension in the direction parallel to the long axis of the useful scan area shall not exceed 0.060 inch (1.52 mm). Its maximum dimension in a direction parallel to the short axis of the useful scan area shall not exceed 0.040 inch (1.02 mm).

6. Grid No. 1 at beam cutoff.

7. The Eis required to correct for barrel and pincushion distortion shall be within the specified limits.

8. Spot size shall be measured with the spot stationary at the center of the tube screen, with grid No. 1 adjusted between 5 and 6 volts above visual spot cutoff and with focus and astigmatism voltages adjusted for smallest round spot. The diameter of the spot, measured with a suitable optical instrument, shall not exceed the specified amount.

9. If these tests have previously been performed by the tube manufacturer, and if the material, manufacturing process, and test requirements are the same as when the tests were performed, the subject test(s) need not be performed.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T15:12Z
Check the source to verify that this is the current version before use.

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Army - EL, MI
Navy -
Air Force - 11, 17, 80
DSA - ES

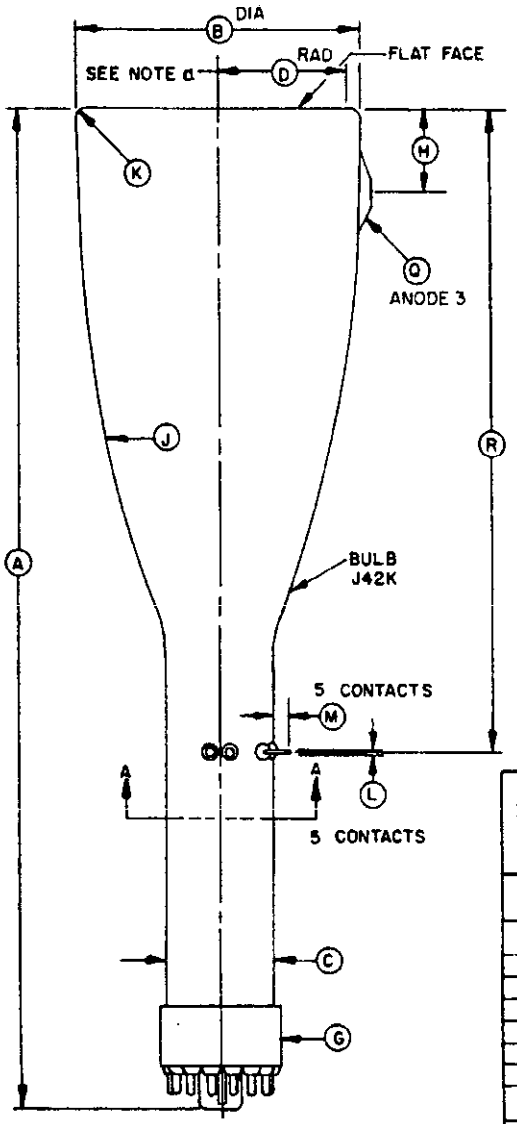
User activities:
Army - ME, WC
Navy - AS, OS, MC, CG, SH
Air Force - 19

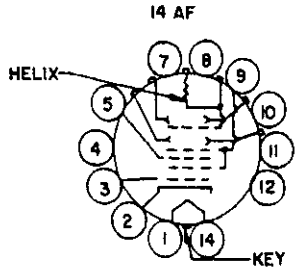
Preparing activity: Army - EL

Agent: DSA - ES

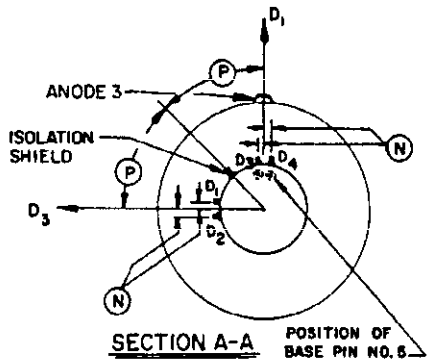
(Project 5960-2443)

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T15:12Z
Check the source to verify that this is the current version before use.





PIN CONNECTIONS	
Pin No.	Element
1	heater
2	cathode
3	grid No. 1
4	nc
5	anode 1
7	nc
8	nc
9	anode 2
10	nc
11	nc
12	nc
14	heater



Ltr	Dimensions in inches with metric equivalents (mm) in parentheses		Notes
	Minimum	Maximum	
Quality conformance inspection, part 2			
A	17.312 (439.70)	17.688 (449.30)	c
C	1.938 (49.21)	2.063 (52.39)	
D	2.250R (57.20)		
H	1.250 (31.80)	1.500 (38.10)	
L	.038 (.97)	.042 (1.07)	
M	.172 (4.37)	.203 (5.16)	
P	40°	50°	
R	11.563 (293.70)	11.813 (300.00)	
Quality confromance inspection, part 3 (periodic check)			
B	5.156 (131.00)	5.344 (135.70)	
G	Base: B12-37		
Q	Bulb contact: J1-21		
	Reference dimensions		b
J	27.813 (706.40)R		
K	.250 (6.40)R		
N	.188 (4.80)		

NOTES:
a. Useful screen radius.
b. Dimensions without tolerances are for information and are not required for inspection purposes.

FIGURE 1. Outline drawing of electron tube type 5BGP2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T15:12Z
Check the source to verify that this is the current version before use.